



Time Series Analysis

Seminar 2: ARMA models



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Acronyms used in this material

- ACF: Autocorrelation Function
- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model)
- ARMA: AutoRegressive Moving Average (model)
- BET: Bucharest Exchange Trading (index)
- BIC: Bayesian Information Criterion
- BNR: Banca Națională a României (the National Bank of Romania)
- EODHD: EOD Historical Data (market-data provider)
- GDP: Gross Domestic Product
- HICP: Harmonised Index of Consumer Prices
- MA: Moving Average
- MAE: Mean Absolute Error
- MLE: Maximum Likelihood Estimation
- PACF: Partial AutoCorrelation Function
- RMSE: Root Mean Squared Error
- US: United States
- WN: White Noise

Today's question and route

- **Question:** how do we choose, estimate, check and use an ARMA model for a stationary series?
 - ▶ this seminar comes **before** Lecture 2: the section “What you need today” gives every formula the tasks use
- **Route**
 - ▶ Part A: AR(1), AR(2), MA(1) and ARMA(1,1) on paper; Yule–Walker; AIC and BIC from given log-likelihoods
 - ▶ Part B: the Box–Jenkins steps on US and Romanian GDP growth, and on BET and EUR/RON returns
 - ▶ Part C: an out-of-sample test for Romanian inflation and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	AR(1): moments and forecasts; AR(2): roots and autocorrelations	Solved, Proposed	A1
A3, A4	MA(1) invertibility; ARMA(1,1) weights and a common factor	Solved, Proposed	A3
A5, A6	Yule–Walker and information criteria; criteria and Ljung–Box degrees of freedom	Solved, Proposed	A5
B1, B2	Box–Jenkins: US real GDP growth; Romanian real GDP growth	Solved, Proposed	B1
B3, B4	AR(1) for daily returns: BET; EUR/RON	Solved, Proposed	B3
C1, C2	does an AR beat simple forecasts of inflation? what is wrong in an AI answer?	Proposed	B1, A1

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
US real GDP	statsmodels macrodata	quarterly	1959–2009
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, seasonally adjusted	2000–2026
BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026

- Growth rates in %: $400 \Delta \ln Y_t$ (US, annualised), $100 \Delta \ln Y_t$ (Romania, quarter on quarter); daily log returns $100 \Delta \ln P_t$
- 12-month inflation: $100 (\ln P_t - \ln P_{t-12})$
- In the notebook: `fit_arma(x, p, q)`, `box_jenkins(y, name)`, `b3_returns('bet')`; no account or key is needed

What you need today (1/4): the models

- $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ white noise; lag operator $LX_t = X_{t-1}$ (Seminar 1)
 - ▶ **AR(p)**: $X_t = c + \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \varepsilon_t$, i.e. $\phi(L)X_t = c + \varepsilon_t$, $\phi(z) = 1 - \phi_1 z - \dots - \phi_p z^p$
 - ▶ **MA(q)**: $X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} = \mu + \theta(L)\varepsilon_t$
 - ▶ **ARMA(p, q)**: $\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$
- **Stationary (causal)**: all roots of $\phi(z) = 0$ satisfy $|z| > 1$; then $X_t = \mu + \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$
 - ▶ AR(1): $|\phi| < 1$; AR(2): $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, $|\phi_2| < 1$
- **Invertible**: all roots of $\theta(z) = 0$ satisfy $|z| > 1$; MA(1): $|\theta| < 1$
 - ▶ then $\varepsilon_t = \sum_j \pi_j X_{t-j}$; MA(1): $\pi_j = (-\theta)^j$

What you need today (2/4): moments

- **AR(1):** $\mu = c/(1 - \phi)$, $\gamma(0) = \sigma^2/(1 - \phi^2)$, $\rho(h) = \phi^h$, $\psi_j = \phi^j$
 - ▶ half-life of a shock: $\ln 0.5 / \ln \phi$
- **AR(2), Yule–Walker:** $\rho(1) = \phi_1/(1 - \phi_2)$, $\rho(h) = \phi_1\rho(h-1) + \phi_2\rho(h-2)$, $h \geq 2$
 - ▶ complex roots when $\phi_1^2 + 4\phi_2 < 0$: the ACF is a damped wave
- **MA(1):** $\gamma(0) = \sigma^2(1 + \theta^2)$, $\gamma(1) = \theta\sigma^2$, $\rho(1) = \theta/(1 + \theta^2)$, $\rho(h) = 0$ for $h \geq 2$
 - ▶ θ with σ^2 and $1/\theta$ with $\theta^2\sigma^2$ have the same autocovariances
- **ARMA(1,1)** $X_t = \phi X_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1}$: $\psi_j = \phi^{j-1}(\phi + \theta)$, $j \geq 1$
 - ▶ $\gamma(0) = \sigma^2 \frac{1+2\phi\theta+\theta^2}{1-\phi^2}$, $\rho(1) = \frac{(1+\phi\theta)(\phi+\theta)}{1+2\phi\theta+\theta^2}$, $\rho(h) = \phi\rho(h-1)$

What you need today (3/4): identification and estimation

- ▣ **Identification:** $AR(p)$: PACF cuts off after p ; $MA(q)$: ACF cuts off after q ; $ARMA$: both decay
 - ▶ band $\pm 1.96/\sqrt{T}$ for the sample ACF and PACF
- ▣ **Yule–Walker** for $AR(2)$, with sample autocorrelations $\hat{\rho}_1, \hat{\rho}_2$:
 - ▶ $\hat{\phi}_1 = \hat{\rho}_1(1 - \hat{\rho}_2)/(1 - \hat{\rho}_1^2)$, $\hat{\phi}_2 = (\hat{\rho}_2 - \hat{\rho}_1^2)/(1 - \hat{\rho}_1^2)$, $\hat{\sigma}^2 = \hat{\gamma}(0)(1 - \hat{\phi}_1\hat{\rho}_1 - \hat{\phi}_2\hat{\rho}_2)$
- ▣ **Maximum likelihood (MLE):** the default of `statsmodels ARIMA(x, order=(p, 0, q))`
 - ▶ const in the output is the mean μ ; standard errors give t -ratios as in regression
- ▣ **Information criteria**, k = number of parameters $(\phi, \theta, \mu, \sigma^2)$:
 - ▶ $AIC = -2 \ln L + 2k$, $BIC = -2 \ln L + k \ln T$; the smallest value wins; BIC prefers smaller models

What you need today (4/4): diagnostics and forecasts

- **Ljung–Box on residuals:** $Q^*(m) = T(T+2) \sum_{h=1}^m \hat{\rho}_{\hat{\varepsilon}}(h)^2 / (T-h) \approx \chi^2(m-p-q)$
 - ▶ 5% critical values: $\chi_{0.95}^2(5) = 11.07$, $\chi_{0.95}^2(7) = 14.07$, $\chi_{0.95}^2(8) = 15.51$, $\chi_{0.95}^2(9) = 16.92$, $\chi_{0.95}^2(10) = 18.31$
 - ▶ **Jarque–Bera:** $\frac{T}{6}(S^2 + (K-3)^2/4) \approx \chi^2(2)$, S skewness, K kurtosis; critical value 5.99
- **Forecasts:** replace future shocks by 0 and future values by their forecasts
 - ▶ AR(1): $\hat{X}_{T+h} = \mu + \phi^h(X_T - \mu)$; MA(q): $\hat{X}_{T+h} = \mu$ for $h > q$
 - ▶ error variance $\sigma_h^2 = \sigma^2(1 + \psi_1^2 + \dots + \psi_{h-1}^2)$; 95% interval $\hat{X}_{T+h} \pm 1.96 \sigma_h$
- **RMSE** (root mean squared error) $= \sqrt{\frac{1}{n} \sum e_i^2}$; **MAE** (mean absolute error) $= \frac{1}{n} \sum |e_i|$ ([Chapter 0](#))

A1: an AR(1) process [Solved]

Task

- ▣ $X_t = 1 + 0.6X_{t-1} + \varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, 0.64)$, Gaussian.
- ▣ 1. Check stationarity and compute the mean and $\gamma(0)$.
- ▣ 2. Compute $\rho(2)$, $\rho(3)$ and the half-life of a shock.
- ▣ 3. With $X_T = 4$, forecast X_{T+1} and X_{T+2} .
- ▣ 4. Compute the 95% intervals for both forecasts.
- ▣ Report: eight numbers and one sentence on mean reversion.

Solution

- ▣ 1. $|0.6| < 1$: stationary; $\mu = 1/0.4 = 2.5$;
 $\gamma(0) = 0.64/(1 - 0.36) = 1.00$
- ▣ 2. $\rho(2) = 0.6^2 = 0.36$, $\rho(3) = 0.216$; half-life
 $\ln 0.5 / \ln 0.6 = 1.36$ periods
- ▣ 3. $\hat{X}_{T+1} = 2.5 + 0.6 \cdot 1.5 = 3.40$; $\hat{X}_{T+2} = 2.5 + 0.36 \cdot 1.5 = 3.04$
- ▣ 4. $\sigma_1 = 0.8$: $[1.832, 4.968]$; $\sigma_2^2 = 0.64(1 + 0.36) = 0.8704$,
 $\sigma_2 = 0.933$: $[1.21, 4.87]$
- ▣ The forecasts move from 4 back towards 2.5, and the intervals widen towards ± 1.96 .

A2: an AR(2) process [Proposed]

Solution discussed in class

Task

- ▣ $X_t = 0.7X_{t-1} - 0.1X_{t-2} + \varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A1.
- ▣ 1. Write the model with the lag operator and factorise $\phi(z)$.
- ▣ 2. Find the roots and decide whether the process is stationary; check the three triangle conditions.
- ▣ 3. Compute $\rho(1)$, $\rho(2)$ and $\rho(3)$ from the Yule–Walker equations.
- ▣ 4. Compute ψ_1 , ψ_2 and ψ_3 .
- ▣ Report: the factorisation, two roots and six numbers.



A3: an MA(1) that is not invertible [Solved]

Task

- ▣ $X_t = \varepsilon_t + 2\varepsilon_{t-1}$, $\varepsilon_t \sim \text{WN}(0, 1)$.
- ▣ 1. Compute $\gamma(0)$, $\gamma(1)$ and $\rho(1)$.
- ▣ 2. Say whether the process is stationary and whether it is invertible.
- ▣ 3. Find the invertible MA(1) with the same autocovariances.
- ▣ 4. Write the first four weights of its AR(∞) form and the forecast $\hat{X}_{T+2}$.
- ▣ Report: five numbers, two verdicts and the new model.

Solution

- ▣ 1. $\gamma(0) = 1 + 4 = 5$, $\gamma(1) = 2 \cdot 1 = 2$, $\rho(1) = 2/5 = 0.4$
- ▣ 2. Stationary (every finite MA is); not invertible: the root of $1 + 2z$ is -0.5 , inside the circle
- ▣ 3. $\theta^* = 1/2$, $\sigma^{*2} = 2^2 \cdot 1 = 4$: $\gamma(0) = 4 \cdot 1.25 = 5$, $\gamma(1) = 0.5 \cdot 4 = 2$
- ▣ 4. $u_t = X_t - 0.5X_{t-1} + 0.25X_{t-2} - 0.125X_{t-3} + \dots$;
 $\hat{X}_{T+2} = \mu = 0$ (beyond $q = 1$)

A4: an ARMA(1,1) process [Proposed]

Solution discussed in class

Task

- ▣ $X_t = 0.5X_{t-1} + \varepsilon_t + 0.3\varepsilon_{t-1}$, $\varepsilon_t \sim \text{WN}(0, 1)$. Model: A1, A3.
- ▣ 1. Check stationarity and invertibility.
- ▣ 2. Compute ψ_1 , ψ_2 and ψ_3 .
- ▣ 3. Compute $\gamma(0)$, $\rho(1)$ and $\rho(2)$.
- ▣ 4. Show that $X_t = 0.5X_{t-1} + \varepsilon_t - 0.5\varepsilon_{t-1}$ is white noise, and say what an estimation program would report for it.
- ▣ Report: two verdicts, six numbers and one sentence.

A5: Yule–Walker and information criteria [Solved]

Task

- ▣ A stationary series has $T = 100$, $\hat{\gamma}(0) = 4$, $\hat{\rho}(1) = 0.75$, $\hat{\rho}(2) = 0.65$.
- ▣ 1. Estimate an AR(2) by Yule–Walker: $\hat{\phi}_1$, $\hat{\phi}_2$, $\hat{\sigma}^2$.
- ▣ 2. Check that the estimated model is stationary.
- ▣ 3. Maximum likelihood gives $\ln L = -214.6$ for AR(1) ($k = 3$) and -210.9 for AR(2) ($k = 4$); compute AIC and BIC.
- ▣ 4. Choose a model with each criterion.
- ▣ Report: three estimates, four criteria and a choice.

Solution

- ▣ 1. $1 - 0.75^2 = 0.4375$; $\hat{\phi}_1 = 0.75 \cdot 0.35 / 0.4375 = 0.6$;
 $\hat{\phi}_2 = (0.65 - 0.5625) / 0.4375 = 0.2$
- ▣ $\hat{\sigma}^2 = 4(1 - 0.6 \cdot 0.75 - 0.2 \cdot 0.65) = 1.68$
- ▣ 2. $\hat{\phi}_1 + \hat{\phi}_2 = 0.8 < 1$, $\hat{\phi}_2 - \hat{\phi}_1 = -0.4 < 1$, $|\hat{\phi}_2| < 1$: stationary
- ▣ 3. $\ln 100 = 4.605$; AR(1): AIC = 435.2, BIC = 443.02; AR(2): AIC = 429.8, BIC = 440.22
- ▣ 4. Both criteria choose AR(2): the gain in $\ln L$ (3.7) exceeds both penalties per parameter (1 and $\ln(100)/2$).

A6: four candidates and the Ljung–Box test [Proposed]

Solution discussed in class

Task

- ▣ $T = 120$; MLE gives $\ln L = -250.3$ (AR(1), $k = 3$), -247.1 (AR(2), $k = 4$), -247.6 (ARMA(1,1), $k = 4$), -246.9 (ARMA(2,1), $k = 5$). Model: A5.
- ▣ 1. Compute AIC and BIC for the four models.
- ▣ 2. Choose a model with each criterion.
- ▣ 3. The residuals of AR(2) give $Q^*(10) = 14.2$; find the degrees of freedom and decide at 5%.
- ▣ 4. Explain what changes if one uses 10 degrees of freedom.
- ▣ Report: a table of eight numbers, two choices and one decision.



B1: US real GDP growth [Solved]

- ▣ **Question:** which ARMA model describes US quarterly real GDP growth, and how persistent are growth shocks?
- ▣ **Inputs:** statsmodels macrodata, real GDP, 1959Q2–2009Q3; $y_t = 400 \Delta \ln Y_t$ (annualised %)
- ▣ **Tasks**
 1. Plot y_t and its ACF and PACF up to lag 12, and propose candidate models.
 2. Estimate white noise, AR(1), AR(2), MA(2) and ARMA(1,1) by maximum likelihood, and tabulate AIC and BIC.
 3. Test the residuals of the BIC model with Ljung–Box $Q^*(8)$ on $8 - p - q$ degrees of freedom, and with Jarque–Bera.
 4. Interpretation: how many quarters does it take for half of a growth shock to fade?
- ▣ **Report:** the chart, a table of ten criteria, two test results and one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



	WN	AR(1)	AR(2)	MA(2)	ARMA(1,1)
AIC	1084.56	1066.98	1063.70	1065.59	1065.02
BIC	1091.17	1076.91	1076.93	1078.83	1078.25

- $T = 202$, ACF 0.30, 0.24, 0.09 and PACF 0.30, 0.16, -0.02 (band ± 0.14): AR(1) or AR(2)
- BIC: AR(1), $\hat{\mu} = 3.12$, $\hat{\phi} = 0.306$ (0.060); AIC: AR(2), $\hat{\phi}_2 = 0.163$ (0.062); the BIC values differ by 0.02
- AR(1) residuals: $Q^*(8) = 11.08$ on 7 df, $p = 0.135$; Jarque–Bera 16.0, $p < 0.001$
- Interpretation: half-life $\ln 0.5 / \ln 0.306 = 0.59$ quarters: growth shocks are short-lived; output levels, not growth rates, carry the persistence

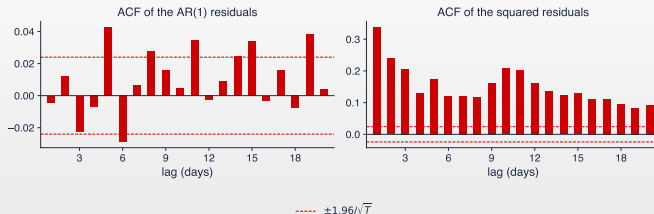
B2: Romanian real GDP growth [Proposed]

- **Question:** does Romanian quarterly real GDP growth have any ARMA structure?
- **Inputs:** Eurostat, chain-linked volumes, seasonally and calendar adjusted, 2000Q1–2026Q2; $y_t = 100 \Delta \ln Y_t$; model: B1
- **Tasks**
 1. Repeat the four steps of B1 for y_t (`ro_gdp_growth('qoq')`).
 2. Run Ljung–Box $Q^*(8)$ on the series itself.
 3. Repeat the analysis without the four quarters of 2020, and compare.
 4. Interpretation: why is the best forecast of next quarter's growth simply the mean?
- **Report:** the chart, a table of ten criteria, the tests and two sentences
- **Notebook:** section B2

B3: BET daily returns [Solved]

- ▣ **Question:** is the first-order autocorrelation of BET returns useful for forecasting?
- ▣ **Inputs:** BET closes since 2000 (EODHD); $r_t = 100 \Delta \ln P_t$, $T = 6\,670$, last day 18 September 2026
- ▣ **Tasks**
 1. Estimate an AR(1) by maximum likelihood and report $\hat{\phi}$, its standard error and t -ratio, and R^2 .
 2. Test the residuals with Ljung–Box $Q^*(10)$ on 9 degrees of freedom, and the squared residuals with $Q^*(10)$.
 3. Compute the kurtosis and the Jarque–Bera statistic of the residuals.
 4. Forecast tomorrow's return with its 95% interval.
 5. Interpretation: is the AR(1) coefficient statistically significant, and is it economically useful?
- ▣ **Report:** six numbers, the chart, a forecast with its interval and two sentences
- ▣ **Notebook:** section B3

B3: solution [Solved]



- $\hat{\phi} = 0.109$ (se 0.0052), $t = 20.9$; $R^2 = 1.2\%$; $\hat{\sigma} = 1.390\%$
- Residuals: $Q^*(10) = 29.7$, $p < 0.001$ (some correlation remains); squares: $Q^*(10) = 2496$; kurtosis 14.0, Jarque–Bera 34128
- Last return 2.17%: forecast 0.294%, 95% interval $[-2.43, 3.02]$ (Normal, too narrow in the tails)
- Interpretation: highly significant, but it explains about 1% of the variance and the forecast is tiny relative to the interval; the large Q^* of the squares points to volatility models ([Chapter 5](#))

B4: EUR/RON daily returns [Proposed]

- **Question:** does the EUR/RON exchange rate behave like the BET?
- **Inputs:** BNR reference rate since July 2005, $T = 5\,352$ returns; model: B3
- **Tasks**
 1. Repeat the four steps of B3 (`b3_returns('eurron')`).
 2. Compare $\hat{\phi}$, R^2 and the kurtosis with those of the BET.
 3. Interpretation: why can a managed exchange rate show a larger first-order autocorrelation than a stock index?
- **Report:** a table (two series, six numbers each) and two sentences
- **Notebook:** section B4

C1: does an AR model beat simple forecasts of inflation? [Proposed]

- ▣ **Question:** is an AR(2) better than the historical mean and the last value at forecasting Romanian 12-month inflation one year ahead?
- ▣ **Inputs:** HICP inflation since 2005 (Eurostat); forecast origins every month from January 2015; models: B1, A1
- ▣ **Tasks**
 1. At each origin, estimate an AR(2) on all data up to that month and forecast 12 months ahead.
 2. Compute the same forecasts from the historical mean and from the last observed value.
 3. Compare RMSE and MAE of the three methods, over the full period and before July 2021.
 4. Interpretation: why do all three methods fail in 2022?
- ▣ **Report:** a table of RMSE and MAE, the chart and a plan for a project
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant about the ARMA models of this seminar. The answer:
- ▣ (a) The process $X(t) = 0.6 X(t-1) + 0.5 X(t-2) + e(t)$ is stationary because both coefficients are below 1.
- ▣ (b) An MA(1) with $\theta = 2$ is not stationary, because $|\theta| > 1$.
- ▣ (c) The AR(1) for BET returns has $t = 21$, so yesterday's return explains most of today's return.
- ▣ (d) For the residuals of an ARMA(1,1), Ljung-Box Q(8) is compared with the chi-square distribution with 8 degrees of freedom.
- ▣ (e) The forecast of an MA(2) three steps ahead equals the mean of the series.
- ▣ (f) In an AR(1), $X(t) = c + \phi X(t-1) + e(t)$, the mean of the series is c .
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- Stationarity is read from the roots of $\phi(z)$, invertibility from the roots of $\theta(z)$, never from single coefficients
- The mean of an AR model is $c/\phi(1)$, not the intercept c
- Box–Jenkins: ACF and PACF propose, AIC and BIC choose, residual tests check, forecasts are compared with simple benchmarks
- Ljung–Box on residuals uses $m - p - q$ degrees of freedom
- Statistically significant is not economically large: compare $\hat{\phi}^2$ with the forecast interval

After the seminar

- Lecture 2 derives today's formulas: roots, invertibility, ψ weights, estimators, information criteria, forecast intervals, the Box–Jenkins case of Romanian GDP
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: ARMA forecasts of inflation in Romania and the region against simple benchmarks
- Reading: [Huang and Petukhina \(2022\)](#), Ch. 3–4; [Hyndman and Athanasopoulos \(2021\)](#), Ch. 9; [Brockwell and Davis \(2016\)](#), Ch. 3
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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